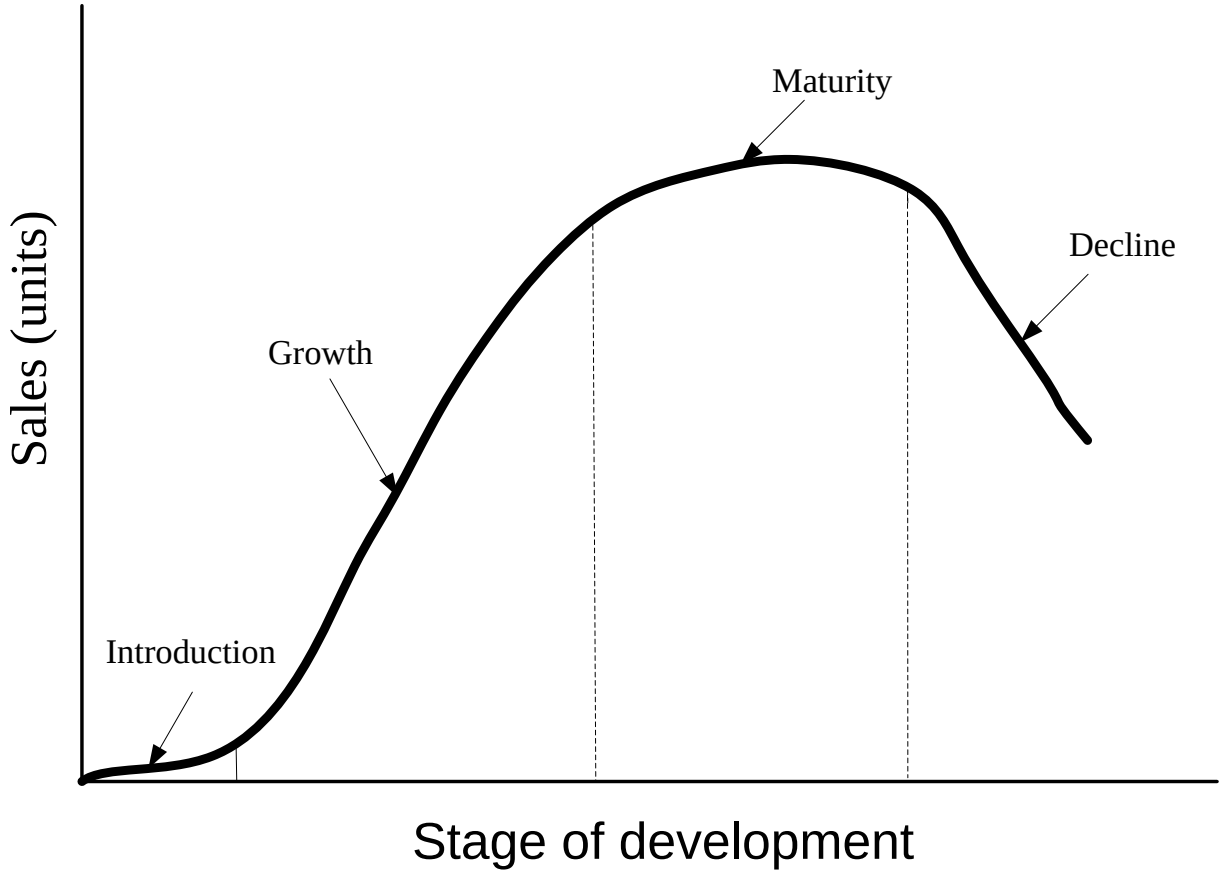
PRODUCT DESIGN
Part-1

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Product Life Cycle



Three important stages of design

1. Conceptual design stage
2. Embodiment stage
3. Detailed design stage

Conceptual Design

- Takes the design problem statement and generates broad solution for it.
- Solution is in the form of design concepts

Embodiment stage

- Concepts are converted into bodily form
- Final check is made on function, spatial compatibility, design aesthetics and economics

Detailed Design

- Outcome of this stage is a set of drawings obtained by rigorous design calculations

Need for conceptual design methodologies

- Not all design criteria can be established with mathematical precision

Concept Selection technique

- Concept evaluation matrix compares generated design concepts against criterion datum.
- For concepts better than datum concept, ‘+’ symbol is assigned; for worse concepts ‘-’ symbol is assigned. ‘s’ indicates datum level concept.
- Best concept will be one with maximum ‘+’ symbols.

PUGH Matrix: Developed by Stuart Pugh in 1960s

Stuart Pugh (1929–1993) was a British Product Designer.

"Total Design is the systematic activity necessary, from the identification of the market/user need, to the selling of the successful product to satisfy that need – an activity that encompasses product, process, people and organization." – Stuart Pugh

A useful site:
https://www.modernanalyst.com/Careers/InterviewQuestions/tabid/128/ID/2159/What-is-a-Pugh-Matrix.aspx

Pugh Matrix: Two Alternatives

Criteria	Weight	Alternative 1 (Baseline)	Alternative 2	Score for Alternative 2
Cost	3	0	-1	-3
Performance	4	0	1	4
Reliability	2	0	0	0
Ease of Use	2	0	1	2
Maintenance	3	0	-1	-3
Total Score				0

Pugh Matrix: Three Alternatives

Criteria	Weight	Alternative 1 (Baseline)	Alternative 2	Score for Alternative 2	Alternative 3	Score for Alternative 3
Cost	3	0	-1	-3	1	3
Performance	4	0	1	4	0	0
Reliability	2	0	0	0	1	2
Ease of Use	2	0	1	2	-1	-2
Maintenance	3	0	-1	-3	1	3
Total Score				0		6

Concept Selection technique

- Main drawback is the large number of combinations possible (${}^{n}C_{2}$)

(Example: There are 10 alternatives. If you compare two at a time, you need ${}^{10}C_{2}$ =10×9/2=45 comparisons.

- Ruthlessness required in eliminating options for quick arrival at preferred solution.

Morphological Analysis

- Problem is decomposed into components or sub-problems.
- Each component represents a meaningful and identifiable part of a major problem.

Morphological matrix

- Each axis represents a major function.
- Each method performing the functions is a co-ordinate along the axis.
- Each cell represents a potential design solution.

Morphological Matrix for Product Design

Functions	Option 1	Option 2	Option 3
Power Source	Battery-operated	Manual (hand pump)	Solar-powered
Heating Mechanism	Electric heater	Chemical packs	External heat source
Water Reservoir	Built-in tank	Detachable tank	No tank (direct feed)
Brewing Method	Drip brewing	Espresso press	Percolation
Material	Stainless steel	Plastic	Aluminum
Size/Portability	Pocket-size	Hand-held	Backpack-compatible
Filter System	Paper filter	Metal mesh filter	Capsule/pod system

Morphological box

- Chief advantage is that unusual or innovative practical solutions will not be overlooked.

Statistical theory of decision

- Application of mathematical techniques in decision-making.
- Wherever uncertainties are present, theory of probability is also used.

Utility theory

- Developed to assist in selecting among a choice of distinct actions, given the uncertainty in the outcome of each action.
- Each design variable is assigned a function representing utility.
- Combination of values that maximizes overall utility is used.

Utility theory: key terms

- Cardinal Approach: Cardinal utility theory assumes that utility (satisfaction from consuming goods) can be measured numerically, using units called "utils", and that these units can be compared and added up.
- Ordinal Approach: Ordinal utility theory posits that consumers can only rank their preferences for goods and services, rather than quantify the satisfaction derived from them.

Marginal Utility

- Marginal utility=Change in in total utility/change in total quantity
- Example: A machine has one sensor and its total utility 60 utils. If add one more sensor (to measure something else), the total utility may become 80 utils. Marginal unit is 20 utils. Adding one more sensor may make total utility 90. Marginal Utility reduce to 10, a reduction from 20.